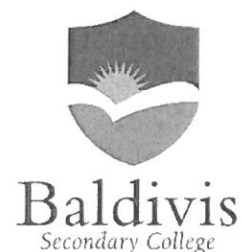


Year 10
Pathway 1

End of Term Test 1



Score: 157 55
Total Time: 60 minutes

Name: _____

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed and the formula page.

1) Expand the following expressions

(5 marks)

a) $x(2x - 3)$

$$2x^2 - 3x \quad \checkmark$$

b) $(x - 6)(x - 2)$

$$x^2 - 8x + 12 \quad \checkmark \quad \text{ms}$$

c) $(x + 4)^2 - 2(x + 1)$

$$\begin{aligned} & \checkmark \quad \checkmark \\ & x^2 + 8x + 16 - 2x - 2 \\ & = x^2 + 6x - 14 \quad \checkmark \end{aligned}$$

2) Simplify the following expressions (leave your answers in index form)

(4 marks)

a) $(2x^5y)^3$

$$8x^{15}y^3 \quad \checkmark \checkmark$$

b) $\frac{3p^2q^5 \times 2p}{15p^6q^5}$

$$\frac{2}{5p^3} \quad \checkmark \checkmark$$

3) Simplify the following expression

(5 Marks)

a) $\frac{x}{5} + \frac{2x}{5}$ $\frac{3x}{5}$ ✓

b) $\frac{t}{8} \times \frac{4t}{3}$ $\frac{4t^2}{24} = \frac{t^2}{6}$ ✓

c) $\frac{2p+1}{3} \div \frac{p-3}{6}$ $\frac{2p+1}{3} \times \frac{6}{p-3} = \frac{4p+2}{p-3}$ ✓✓

4) Factorise the following expressions

2 (3 marks)

a) $x^2 + 7x - 30$ $(x+10)(x-3)$ ✓

b) $f^2 - 100$ $(f-10)(f+10)$

5) Evaluate the following

(5 marks)

a) $36^{\frac{1}{2}}$ $= \sqrt{36} = 6$ ✓

b) $125^{\frac{2}{3}}$ $= (3\sqrt[3]{125})^2 = 5^2 = 25$ ✓✓

c) $\log_4 16$ $= 2$ ✓

d) $\log_2 128$ $= 7$ ✓

6) Simplify the following

(5 marks)

a) $7\sqrt{2} + 9\sqrt{2} - 3\sqrt{2} = 13\sqrt{2}$ ✓

b) $10\sqrt{2} \times 3\sqrt{8} = 30 \times \sqrt{16} = 30 \times 4 = 120$ ✓✓

c) $\frac{7\sqrt{44}}{14\sqrt{11}} = \frac{7 \times \sqrt{4} \times \sqrt{11}}{14\sqrt{11}} = 1$ ✓✓

7)

(6 marks)

a) Write $4^2 = 16$ in log form

$\log_5 25 = 2$ ✓

b) Write $\frac{1}{2} = \log_{49} 7$ in index form

$49^{1/2} = 7$ ✓

c) Evaluate the following

i. $\log_{10} 25 + \log_{10} 4$

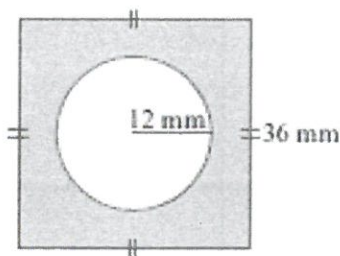
$\log_{10}(100) = 2$ ✓✓

ii. $\log_3 27 + \log_3 2 - \log_3 6$

$= \log_3(54) - \log_3(6)$ ✓✓
 $= \log_3(9) = 2$

8) Find the area of the following shape

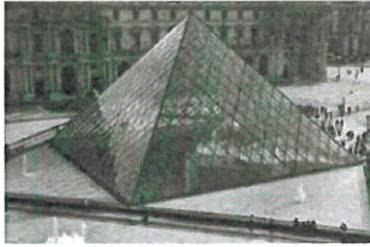
(2 marks)



$36^2 - \pi \times 12^2$ ✓
 $= 843.61 \text{ mm}^2$ ✓

- 9) The main entrance to the Louvre Museum in Paris, France, has been constructed as a pyramid with a square base of length 35m. Each identical triangular face has a height of 27m and is made of glass segments. Find the total surface area of the glass walls of the pyramid.

(2 marks)



$$4 \times \frac{1}{2} \times 35 \times 27 = 1890 \text{ m}^2$$

- 10) Find the surface area and volume for the shapes given below.

10
(11 marks)

Shape	Surface Area	Volume
	$2(11 \times 7 + 11 \times 3 + 3 \times 7) = 262 \text{ cm}^2$	$11 \times 7 \times 3 = 231 \text{ cm}^3$
	$4.2 \times 2.9 + (4.2 \times 2.9 + 5.1 \times 3.4) = 53.66 \text{ m}^2$	$\frac{1}{2} (2.9 \times 4.2) \times 3.4 = 20.706 \text{ m}^3$
	$\pi r s + \pi r^2$ $s = 13$ $\pi \times 5 \times 13 + \pi \times 25 = 90\pi = 282.74 \text{ cm}^2$	$\frac{1}{3} \pi \times 5^2 \times 12 = 100\pi = 314.16 \text{ cm}^3$
	$4\pi r^2$ $r = 13$ $= 2704\pi = 8484.94 \text{ cm}^2$	$\frac{4}{3} \pi r^3 = 73622.18 \text{ cm}^3$

11) Western Wheat has silos for storing wheat. There are two rows of 8 silos that have a cylindrical base (height 6m; diameter 4m) with a dome top (diameter 4m).

(9 marks)

a) What is the surface area of the silos?

$$r=2 \quad 2\pi r \times 2 \times 6 + 2\pi r^2 \quad \checkmark \quad \checkmark$$
$$= 32\pi = 100.53\text{m}^2 \text{ each.}$$

$$16 \times 32\pi = 512\pi$$
$$= 1608.50\text{m}^2. \quad \checkmark$$

b) How much wheat can be stored altogether in these silos?

$$\pi r^2 h + \frac{2}{3} \pi r^3 = 92.15\text{m}^3 \quad \checkmark \quad \checkmark$$

$$\times 16 = 1474.45\text{m}^3 \quad \checkmark$$

c) The wheat is emptied into trucks with rectangular containers 2.4m wide, 5m long and 2.5m high.
How many truckloads are needed to empty all of the silos.

$$1474.45 / 30\text{m}^3 = 49.1 \quad \checkmark$$

$\therefore 50$ required $\checkmark$

End of Test

